



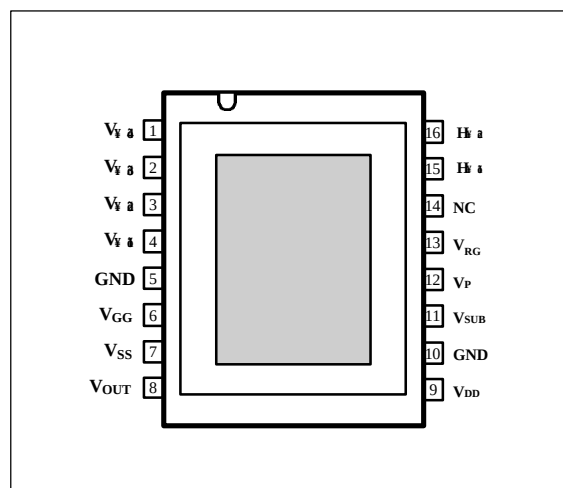
Description

The Ai325 is a 250K pixels CCD area sensor for EIA 1/3inch video cameras. Buried photodiode and micro lens are adopted for low noise, low smear and high sensitivity. This chip also features a strong anti-blooming and electronic shutter with variable charge-storage time.

Feature

- Micro Lens for high sensitivity
- Image-lag is negligible and excellent blooming suppression is performed.
- TTL level(5V) operation on HCCD & RG electrodes.
- 16 pin plastic-DIP.
- Variable electronic shutter of 1/60 to 1/100,000 sec.
- High sensitivity and low smear.

Pin Configuration

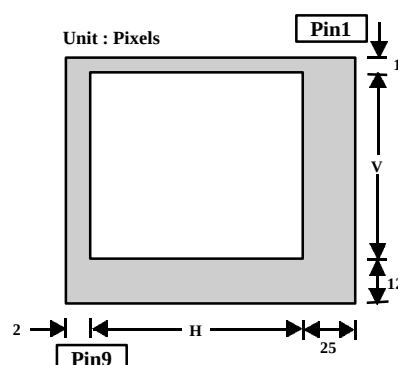


16 Pin Plastic - DIP
(Top View)

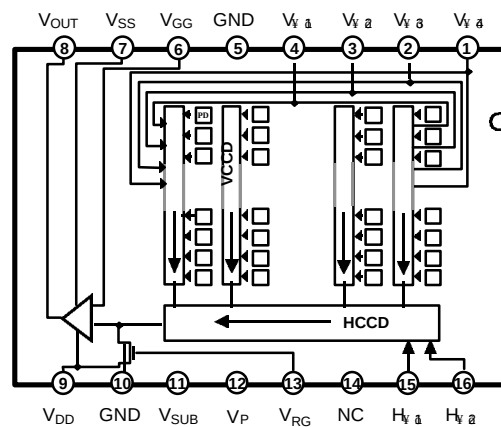
Structure

- Architecture : IT - CCD
- Optical size : 1/3 inch format
- Chip size : 6.0(H) x 5.2(V) $\pm$
- Number of effective pixels :
510 (H) x 492 (V) about 250K pixels
- Number of total pixels :
537 (H) x 505 (V) about 270K pixels
- Pixel size : 9.65 (H) x 7.5 (V) μ m²
- Optical black area
Horizontal direction : Front 2 pixels Rear 25 pixels
Vertical direction : Front 12 pixels Rear 1 pixels
- Number of dummy bits
Horizontal : 16
Vertical : 1 (Even field only)

Optical black position(Top View)



Block Diagram



Pin Description

No.	Symbol	Description	No.	Symbol	Description
1	V_{1a}	Vertical register transfer clock 4	9	V_{DD}	Output amplifier drain bias
2	V_{1b}	Vertical register transfer clock 3	10	GND	Ground
3	V_{2a}	Vertical register transfer clock 2	11	V_{SUB}	Substrate(Overflow drain)bias
4	V_{2b}	Vertical register transfer clock 1	12	V_P	Protection bias
5	GND	Ground	13	V_{RG}	Reset gate clock
6	V_{GG}	Output amplifier gate bias	14	NC	No connection
7	V_{SS}	Output amplifier source bias	15	H_{1a}	Horizontal register transfer clock 1
8	V_{OUT}	CCD Output signal	16	H_{1b}	Horizontal register transfer clock 2

Absolute Maximum Ratings

Parameter	Symbol	Value	Unit
Substrate voltage	$V_{SUB} - GND$	-0.3 to +55	V
Supply voltage	$V_{DD}, V_{OUT}, V_{SS} - GND$	-0.3 to +18	V
	$V_{DD}, V_{OUT}, V_{SS} - V_{SUB}$	-55 to +10	V
Vertical clock input voltage	$V_{1a}, 2, 3, 4 - GND$	-10 to +20	V
	$V_{1a}, 2, 3, 4 - V_P$	-0.3 to +27	V
	$V_{1a}, 2, 3, 4 - V_{SUB}$	-55 to +10	V
Horizontal clock input voltage	$H_{1a}, H_{1b} - GND$	-10 to +15	V
Between vertical clock input pins	$V_{1a} - V_{1b}$	-10 to +15	V
Between horizontal clock and vertical clock input pins	$H_{1a}, H_{1b} - V_{1a}$	-17 to +17	V
Output pin voltage	$V_{RG}, V_{GG} - GND$	-10 to +15	V
	$V_{RG}, V_{GG} - V_{SUB}$	-55 to +10	V
Protective circuit voltage	$V_P - V_{SUB}$	-65 to 0.3	V
Storage temperature	T_{STG}	-30 to 80	°C
Operation temperature	T_{OPR}	-10 to 60	°C

Bias Condition

Parameter	Symbol	Min.	Typ.	Max.	Unit	Remark
Output amplifier drain voltage	V_{dd}	14.5	15.0	15.5	V	
Output amplifier gate voltage	V_{gg}	1.5	2.0	2.5	V	
Output amplifier source voltage	V_{ss}	Ground through 680Ω Resistor, ±5%				
Substrate voltage adjustment range	V_{sub}	5		15	V	
Fluctuation range after substrate voltage adjustment	ΔV_{sub}	-1		1	V	
Reset gate clock voltage adjustment range	V_{rgl}	0		4	V	
Fluctuation range after reset gate voltage adjustment	ΔV_{rgl}	-3		3	%	
Protection bias	V_p	Set to low level of vertical transfer clock				

DC Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Output amplifier drain current	I_{DD}	-	3	-	μA

Driving Condition

Parameter	Symbol	Min.	Typ.	Max.	Unit
Vertical clock high voltage	V_{H1}, V_{H3}	14.5	15.0	15.5	V
Vertical clock middle voltage	$V_{M1, 2, 3, 4}$	-0.2	0.0	0.2	V
Vertical clock low voltage	$V_{L1, 2, 3, 4}$	-9.0	-8.5	-8.0	V
Horizontal clock high voltage	$V_{H1, 2}$	4.5	5.0	5.5	V
Horizontal clock low voltage	$V_{L1, 2}$	-0.5	0.0	0.5	V
RG clock voltage difference	ΔV_{RGHL}	4.7	5.0	5.3	V
Substrate clock voltage	V_{SUB}	23	24	25	V

Electro-optical Performance

Item	Symbol	Min.	Typ.	Max.	Unit	Measurement Method	Remark
Sensitivity	SENS	55	70		μLux	1	
Saturation signal	V_{SAT}	800			μV	2	Temp=60 $^{\circ}\text{C}$
Smear	S_{MR}			0.015	%	3	
Blooming	BL			1	%	4	
Video signal shading	OSNU			15	%	5	
Dark signal level	V_{DARK}			2	μV	6	Temp=60 $^{\circ}\text{C}$
Dark signal shading	DSNU			2	μV	7	Temp=60 $^{\circ}\text{C}$

Measurement Method

1. Sensitivity

- Set to SILC (Standard Illumination Conditions*)
- Measure the average value of signal output (V_{out})
- Calculate the efficiency of V_{out} to light intensity

2. V_{sat}

- Adjust light intensity to 200 times of SILC
- Measure the average value of signal output

3. Smear

- Adjust light intensity to 200 times of SILC & readout clock
- Measure the signal output at horizontal optical black (V_{hopb})
- Measure the signal output at vertical blanking dummy (V_{vbd})
- $\text{Smear} = \{ (V_{vbd} - V_{hopb}) / V_{sat} \} \times 100 (\%)$

4. Blooming

- Adjust light intensity to 200 times of SILC & readout clock
- Measure the signal output at horizontal optical black (V_{hopb})
- Measure the signal output at blooming dummy area (V_{bd})
- $\text{Blooming} = \{ (V_{bd} - V_{hopb}) / V_{sat} \} \times 100 (\%)$

5. OSNU

- Set to SILC
- Measure the average value of signal output (V_{out})
- Measure the maximum value and the minimum value of signal output
- $\text{OSNU} = (V_{max} - V_{min}) / V_{out} \times 100 (\%)$

6. V_{dark}

- Measure the average value of signal output at dark condition

7. DSNU

- Measure the voltage difference between minimum and maximum of dark signal

* Standard Illumination Conditions

- Measure the average value of output of linear region
- At this time, measure the light intensity of illumination at CCD face plate
- Define SILC with above
- Light source: Tungsten lamp(3100K)
- Use a standard test lens at F8

Notes on Handling

1) Static charge prevention

CCD image sensors are easily damaged by static discharge. Before handling be sure to take the following protective measures.

- a) Either handle bare handed or use non-chargeable gloves, clothes or material. Also use conductive shoes.
- b) When handling directly use an earth band.
- c) Install a conductive mat on the floor or working table to prevent the generation of static electricity.
- d) Ionized air is recommended for discharge when handling CCD image sensor.
- e) For the shipment of mounted substrates, use boxes treated for the prevention of static charges.

2) Soldering

- a) Make sure the package temperature does not exceed 80 °C.
- b) Solder dipping in a mounting furnace causes damage to the glass and other defects. Use a ground 30W soldering iron and solder each pin in less than 2 seconds. For repairs and remount, cool sufficiently.
- c) To dismount an image sensor, do not use a solder suction equipment. When using an electric desoldering tool, use a thermal controller of the zero cross On/Off type and connect it to ground.

3) Dust and Dirt protection

- a) Perform all assembly operations in a clean room (class 1000 or less).
- b) Do not either touch glass plates by hand or have any object come in contact with glass surfaces. Should dirt stick to a glass surface, blow it off with an air blower. (For dirt stuck through static electricity ionized air is recommended.)
- c) Clean with a cotton bud and ethyl alcohol if the grease stained. Be careful not to scratch the glass.
- d) Keep in a case to protect from dust and dirt. To prevent dew condensation, preheat or precool when moving to a room with great temperature differences.

4) Others

- a) Do not expose to strong light (sun rays) for long periods.
- b) Exposure to high temperature or humidity will affect the characteristics. Accordingly avoid storage or usage in such conditions.

Application Circuit

